# 2026 May I
# Module 1

# Question N. 1

An artist made 14 pieces of pottery using 48 pounds of clay. The pottery included only bowls and vases. Each bowl used 3 pounds of clay and each vase used 5 pounds of clay. How many vases did the artist make?

A) 3

B) 6

C) 8

D) 11

# Question N. 2

The equation $2x^2 + 18x + 2y^2 + 16y = \frac{11}{2}$ defines a circle in the $xy$-plane. What is the radius of the circle?

A) $\frac{\sqrt{22}}{2}$

B) $\sqrt{11}$

C) $\sqrt{39}$

D) $\frac{\sqrt{602}}{2}$

# Question N. 3

The equation $E(t) = 3(1.7)^t$ gives the estimated number of employees at a certain business, where $t$ is the number of years since the business opened. Which of the following is the best interpretation of the number 3 in this context?

A) The increase in the estimated number of employees each year

B) The estimated number of employees when the business opened

C) The percent increase in the estimated number of employees each year

D) The number of years the business has been open

# Question N. 4

The table gives the distribution of flavor and type of topping for customer orders at an ice cream shop.

| | **Type of topping** | |
|---|---|---|
| **Flavor of ice cream** | **Sprinkles** | **No Sprinkles** |
| Chocolate | 50 | 40 |
| Vanilla | 30 | 10 |
| Twist | 80 | 20 |

If a customer order is selected at random, what is the probability of selecting an order with sprinkles, given the flavor of ice cream is vanilla?

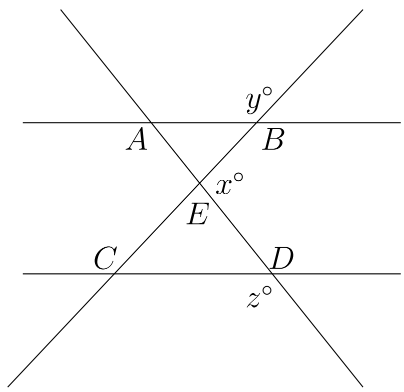
A) 0.13

B) 0.19

C) 0.38

D) 0.75

# Question N. 5




In the figure, line $AB$ is parallel to line $CD$, and line $AD$ intersects line $BC$ at point $E$. If $y = 115$ and $z = 118$, what is the value of $x$?

A) 53

B) 62

C) 127

D) 143

# Question N. 6

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 1900% of the number of prey at the start of the week, and the number of predators had increased by 460% of the number of predators at the start of the week. The number of predators at the end of the week was $p\%$ less than the number of prey at the end of the week. What is the value of $p$?

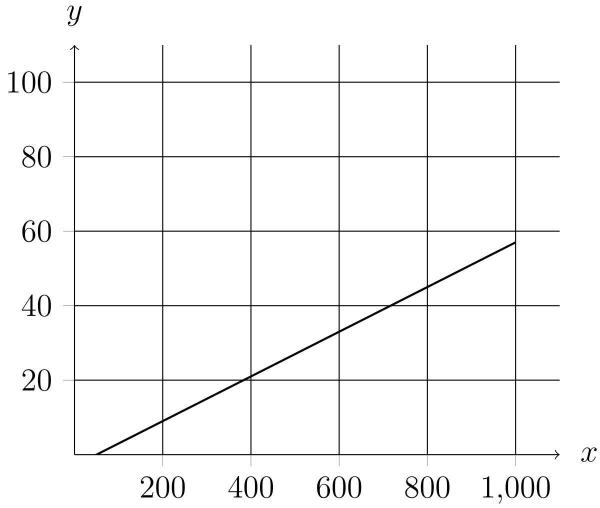
# Question N. 7

$$-3x^2 + bx - 48 = 0$$

In the given equation, $b$ is a positive integer. The equation has no real solution. What is the largest possible value of $b$?

# Question N. 8

The function $f$ models the resistivity of copper, in nanoohm·meters, as a function of the temperature $x$, in kelvins, of the copper. The graph of $y = f(x)$ is shown.




According to the model, which of the following statements about the estimated resistivity of copper is true?

A) It increases at a rate of $\frac{3}{5}$ nanoohm·meters per 100 kelvins.

B) It increases at a rate of 6 nanoohm·meters per 100 kelvins.

C) At 100 kelvins, it is 10 nanoohm·meters.

D) From 200 kelvins to 400 kelvins, it decreases 12 nanoohm·meters.

# Question N. 9

Line $t$ in the $xy$-plane has a slope of $\frac{1}{9}$ and passes through the point $(45, -8)$. Which equation defines line $t$?

A) $y = -13x + \frac{1}{9}$

B) $y = \frac{x}{9} - 13$

C) $y = \frac{x}{9} - 8$

D) $y = 45x - 8$

# Question N. 10

One gallon of paint will cover 280 square feet of a surface. A room has a total wall area of $w$ square feet. Which equation represents the total amount of paint $P$, in gallons, needed to paint the walls of the room twice?

A) $P = 560w$

B) $P = 280w$

C) $P = \frac{w}{140}$

D) $P = \frac{w}{280}$

# Question N. 11

If $|12 - 2x| = 48$, which of the following is a value of $6 - x$?

A) 0

B) 6

C) 24

D) 36

# Question N. 12

| $x$ | $y$ |
|---|---|
| $-\frac{2}{3}$ | 0 |
| 0 | $-48$ |
| 4 | 0 |

The table shows three values of $x$ and their corresponding values of $y$. There is a quadratic relationship between $x$ and $y$. An equation that represents this relationship can be written as $y = 18x^2 - bx - 48$, where $b$ is a constant. What is the value of $b$?

# Question N. 13

Line $p$ in the $xy$-plane contains the points $(11, 0)$ and $(11, 11)$. Which equation defines line $p$?

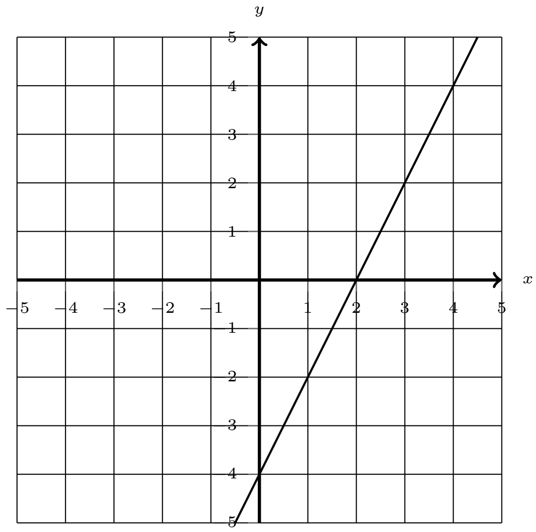
A) $x = 0$

B) $x = 11$

C) $y = 0$

D) $y = 11$

# Question N. 14




The line shown can be represented by the equation $ax + by = 32$, where $a$ and $b$ are constants. Which of the following is true about $a$ and $b$?

A) $a < 0$ and $b < 0$

B) $a < 0$ and $b > 0$

C) $a > 0$ and $b < 0$

D) $a > 0$ and $b > 0$

# Question N. 15

The area of a rectangular region is increasing at a rate of 330 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

A) 0.51

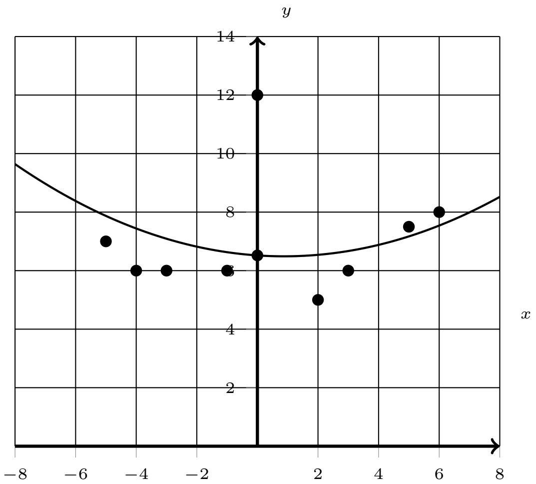
B) 1.68

C) 18.04

D) 30.67

# Question N. 16

The scatterplot shows the relationship between $x$ and $y$ for the 9 data points in data set A.




A quadratic model for the data in data set A can be written as $y = 0.04x^2 - 0.07x + 6.52$. The graph of this model is shown in the $xy$-plane. The data point at $x = 0$ was found to be the result of a recording error and was removed from data set A to create data set B, consisting of the remaining 8 data points. A quadratic model for the data in data set B can be written as $y = ax^2 + bx + c$, where $a$, $b$, and $c$ are constants. If the quadratic models for data sets A and B are calculated in the same way, which of the following statements must be true?

I. $a > 0.04$

II. $c < 6.52$

A) I only

B) II only

C) I and II

D) Neither I nor II

## Question N. 17

Which expression is **NOT** a factor of $1{,}120x^4 - 43{,}750$?

A) $4x^2 + 25$

B) $2x^2 - 5$

C) $2x + 5$

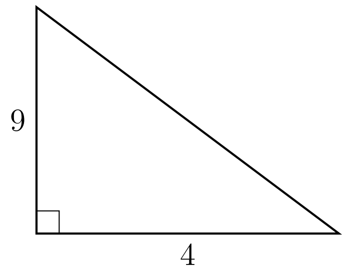
D) $70$

## Question N. 18

$$3 - y = x$$
$$9 + y = x$$

The solution to the given system of equations is $(x, y)$. What is the value of $x$?

# Question N. 19



*Note: Figure not drawn to scale.*

A right triangle has the dimensions shown, in centimeters (cm). What is the area of this triangle, in $cm^2$?

# Question N. 20

$$y = 3x^2 + 28$$
$$y = px + 25$$

In the given system of equations, $p$ is a positive constant. The graphs of the equations in the given system intersect at exactly one point, $(x, y)$, in the $xy$-plane. What is the value of $x$?

A) 1

B) 6

C) 31

D) 53

# Question N. 21

$$-3x + 24px = 96$$

In the given equation, $p$ is a constant. The equation has no solution. What is the value of $p$?

A) 0

B) $\frac{1}{8}$

C) $\frac{4}{3}$

D) 4

# Question N. 22

$$f(x) = (2x - 15)(2x + 7)$$

What is the minimum value of the given function?

A) $-121$

B) $-\frac{7}{2}$

C) 2

D) 8

# Module 2

# Question N. 1

The function $f$ is a quadratic function. In the $xy$-plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 52)$ and $(-1, 181)$. What is the value of $f(-2) - f(0)$?

A) 138

B) 224

C) 344

D) 396

# Question N. 2

What is the $x$-intercept of the graph of $y = \dfrac{28 - x}{ax + b}$ in the $xy$-plane, where $a$ and $b$ are positive constants?

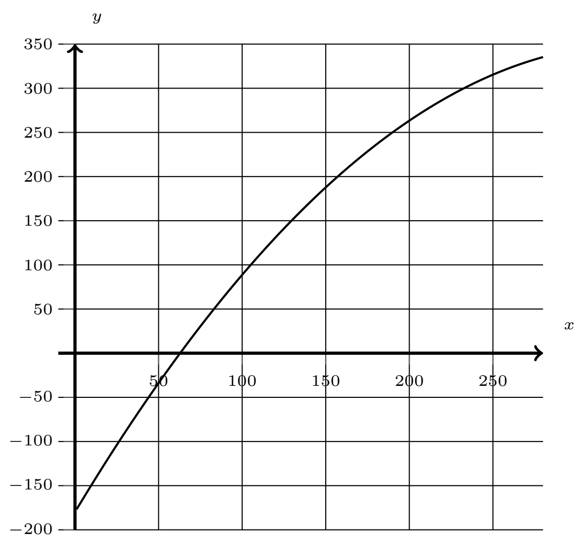
A) $\left(-\dfrac{b}{a}, 0\right)$

B) $\left(0, \dfrac{28}{b}\right)$

C) $(0, -28)$

D) $(28, 0)$

# Question N. 3




The graph shows the estimated boiling point $y$, in degrees Celsius, of a straight-chain alkane with a molecular weight of $x$ grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(169.39, 219.86)$?

A) A straight-chain alkane with a molecular weight of 219.86 grams per mole has an estimated boiling point of 169.39 degrees Celsius.

B) A straight-chain alkane with a molecular weight of 169.39 grams per mole has an estimated boiling point of 219.86 degrees Celsius.

C) The minimum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 169.39 grams per mole and an estimated boiling point of 219.86 degrees Celsius.

D) The maximum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 169.39 grams per mole and an estimated boiling point of 219.86 degrees Celsius.

# Question N. 4

Which expression is equivalent to $\frac{1}{9y^8}$, where $y > 1$?

A) $9y^{-8}$

B) $(9y)^{-8}$

C) $\frac{9}{y^{-8}}$

D) $\frac{y^{-8}}{9}$

# Question N. 5

If $5 + \frac{2(11-x)}{9} = 3(11-x) + 5$, what is the value of $11 + x$?

# Question N. 6

$$\frac{1}{16} = \frac{1}{R_1} + \frac{1}{R_2}$$

The given equation relates the resistances, $R_1$ and $R_2$, in ohms, of two resistors arranged in parallel in a circuit with a total resistance of 16 ohms, where $R_1$ and $R_2$ are positive. Which equation correctly expresses $R_1$ in terms of $R_2$?

A) $R_1 = 16 + R_2$

B) $R_1 = 16 - R_2$

C) $R_1 = \frac{16R_2}{R_2 + 16}$

D) $R_1 = \frac{16R_2}{R_2 - 16}$

# Question N. 7

For two acute angles $X$ and $Y$, the measure of angle $Y$ is $47°$ and $\dfrac{\sin X}{\cos Y} = 1$. What is the measure, in degrees, of angle $X$?

A) 92

B) 47

C) 45

D) 43

# Question N. 8

Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 3.5 micrograms of vitamin B12 are consumed from eating $x$ ounces of cheese and $y$ ounces of tuna, which equation best represents this situation?

A) $0.83x + 0.33y = 3.5$

B) $3.0x + 1.2y = 3.5$

C) $1.2x + 3.0y = 3.5$

D) $0.33x + 0.83y = 3.5$

# Question N. 9

$$f(x) = 11x^{10} + 2x^8$$
$$g(x) = -13x^7 + 7x^5$$

The polynomial $p(x)$ is defined as the product of the given polynomials, $f(x)$ and $g(x)$. What is the coefficient of $x^{15}$ in $p(x)$?

A) 15

B) 51

C) 77

D) 103

# Question N. 10

The price of an item increased by $p\%$ from \$18 to \$21. What is the value of $p$?

## Question N. 11

$$y = 2x^2 - 12x + 20$$
$$y + 3 = 0$$

How many solutions are there to the given system of equations?

A) There is exactly 1 solution.

B) There are exactly 2 solutions.

C) There are exactly 3 solutions.

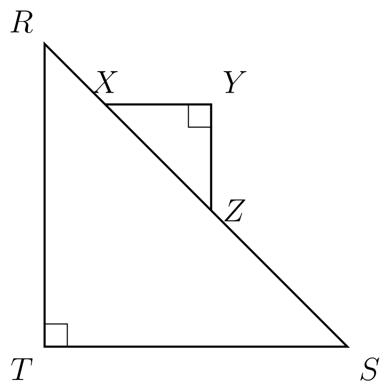
D) There are no solutions.

## Question N. 12

$$(x + 3)(x - 16) = 0$$

What is the positive solution to the given equation?

## Question N. 13




In triangles $RST$ and $XYZ$ shown, $\overline{XY}$ is parallel to $\overline{TS}$ and $\tan R = \frac{160}{231}$. What is the value of $\sin X$ in triangle $XYZ$?

A) $\frac{160}{391}$

B) $\frac{160}{281}$

C) $\frac{231}{281}$

D) $\frac{231}{160}$

# Question N. 14

A bookstore is giving away free books every day. The equation $f(x) = 168 - 12x$ gives the number of free books, $f(x)$, that remain to be given away after $x$ days. Which of the following is the best interpretation of $f(14) = 0$ in this context?

A) The bookstore has 14 free books to give away.

B) The bookstore will give away 14 free books in 12 days.

C) It will take 14 days to give away all the free books.

D) It will take 168 days to give away 14 free books.

# Question N. 15

A right square prism has a height of 9 units. The volume of the prism is 1,521 cubic units. What is the length, in units, of an edge of the base?

A) 9

B) 13

C) 39

D) 169

# Question N. 16

A zoologist observed the nests of wood ducks in an area. Each year, the zoologist recorded the number of eggs in each of the first 35 nests that were observed and created a frequency table for the data set. Which of the following frequency tables represents the data set with the smallest standard deviation?

A)

| Number of eggs | Frequency |
|---|---|
| 10 | 8 |
| 11 | 7 |
| 12 | 5 |
| 13 | 7 |
| 14 | 8 |

B)

| Number of eggs | Frequency |
|---|---|
| 10 | 7 |
| 11 | 7 |
| 12 | 7 |
| 13 | 7 |
| 14 | 7 |

C)

| Number of eggs | Frequency |
|---|---|
| 10 | 2 |
| 11 | 8 |
| 12 | 15 |
| 13 | 8 |
| 14 | 2 |

D)

| Number of eggs | Frequency |
|---|---|
| 10 | 0 |
| 11 | 5 |
| 12 | 25 |
| 13 | 5 |
| 14 | 0 |

# Question N. 17

$$-4x^2 + bx - 16 = 0$$

In the given equation, $b$ is a positive integer. The equation has no real solution. What is the largest possible value of $b$?

# Question N. 18

A community organization surveyed a random sample of residents of the community to estimate the percentage of residents who visit their local library at least once per month. From the survey, the organization estimates that 19% of residents of the community visit their local library at least once per month, with an associated margin of error of 3.7%. The organization repeated the survey with a random sample of residents that was double the original sample size. Assuming the margins of error are calculated in the same way for both studies, which of the following is true about the margin of error associated with the estimate from the larger sample?

A) The margin of error is less than 3.7%.

B) The margin of error is 7.4%.

C) The margin of error is greater than 19%.

D) The margin of error is 38%.

# Question N. 19

Each angle of $\triangle ABC$ is congruent to one of the angles of $\triangle QRS$. If $AB = 56$, $BC = 84$, $AC = 112$, and $RS = 336$, what is one possible value for the perimeter of $\triangle QRS$?

# Question N. 20

The function $f$ is defined by $f(z) = 8z$. Which of the following is equivalent to $f(z-6)$?

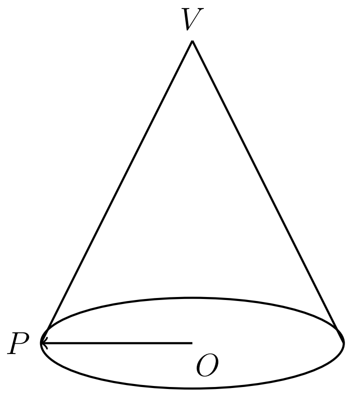
A) $-8z$

B) $8z - 6$

C) $8z - 48$

D) $-8z + 48$

# Question N. 21




For the right circular cone shown, $\overline{OP}$ is a radius of the base and $V$ is the vertex. The cone's lateral surface area, in square meters, is $rL\pi$, where $r$ is the length, in meters, of $\overline{OP}$ and $L$ is the length, in meters, of $\overline{VP}$. The cone has a base area of $81\pi$ square meters and a lateral surface area of $214\pi$ square meters. What is the height of the cone, to the nearest meter?

# Question N. 22

If $x \geq -6$ represents all solutions to the inequality $ax - 16 \leq 14$, where $a$ is a constant, what is the greatest possible value of $a$?

# Answer Sheet

## Module 1

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| $A$ | $C$ | $B$ | $D$ | $C$ | 72 | 23 | $B$ | $B$ | $C$ |

| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|
| $C$ | 60 | $B$ | $C$ | $A$ | $C$ | $B$ | 6 | 18 | $A$ |

| 21 | 22 |
|---|---|
| $B$ | $A$ |

## Module 2

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| $C$ | $D$ | $B$ | $D$ | 22 | $D$ | $D$ | $D$ | $B$ | 50/3 |

| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|
| $D$ | 16 | $C$ | $C$ | 13 | $D$ | 15 | $A$ | 756 | $C$ |

| 21 | 22 |
|---|---|
| 22 | $-5$ |